BT5230: AI for Drug Discovery Mini Project

Multi-Task Prediction of Tox21 Toxicity Assay Activity Using Molecular Structure

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Mini Project Report

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1 Abstract

Early identification of chemical safety signals can reduce time and cost in drug discovery. However, experimental screening across many toxicity-related assays is expensive, while a chemical structure may have labels missing for several assays. This project develops a student-level machine learning workflow for the Tox21 dataset. A compound is represented by an ECFP4 molecular fingerprint and eight simple molecular descriptors. One shared multi-task neural network predicts activity for 12 nuclear-receptor and stress-response assays simultaneously. Missing assay labels are ignored using a masked binary cross-entropy loss. The model is evaluated using a Murcko scaffold split so that closely related chemical cores do not occur in both training and test sets. On the held-out scaffold test set, the multi-task model achieved mean ROC-AUC of 0.741 and mean PR-AUC of 0.342. A conventional set of separate logistic-regression models achieved mean ROC-AUC of 0.716 and mean PR-AUC of 0.307. A Streamlit interface accepts a SMILES string and returns assay probabilities with a similarity-based applicability warning. This tool is for educational early screening only; assay activity is not a direct clinical toxicity prediction.

2 Introduction and Problem Statement

Drug discovery requires many decisions before a molecule can be tested in animals or humans. One important decision is whether a molecule may trigger unwanted biological pathways. Laboratory safety assays are useful but require time, reagents, and money. Therefore, a computational model that screens molecules from their structure can help researchers prioritise compounds.

The problem addressed in this project is:
Given a molecule in SMILES format, predict its probability of being active in each of the 12 Tox21 nuclear-receptor and stress-response assays.

This is a multi-label classification problem because one molecule can be active in more than one assay. It is also a partially labelled problem because the dataset does not contain all 12 assay results for every molecule. The project uses the Tox21 dataset distributed through MoleculeNet [2, 1].

3 Why This Problem Matters

Traditional screening generally tests one compound in one assay at a time. It gives experimental evidence, but it is slow and costly when many candidate molecules must be considered. Structure-based machine learning does not replace experiments. Instead, it provides a first filtering step:

1. A researcher enters or generates candidate SMILES strings.
2. The model ranks assay-activity risk for every candidate.

3. High-risk or uncertain compounds can be prioritised for laboratory testing.

4. Lower-risk candidates can be considered alongside other drug-like properties.

The usefulness of the present approach is that one model gives 12 outputs from one molecular representation. It shares learning across related assays, handles blank labels correctly, and is available through a simple web interface.

4 Objectives

The objectives of this work were:

- Clean and standardise public Tox21 molecular data.
- Create an honest scaffold-based train/validation/test split.
- Train a single multi-task neural network for 12 toxicity-related assays.
- Handle missing labels and class imbalance during training.
- Compare the proposed multi-task model with a traditional single-task logistic-regression baseline.
- Build a Streamlit interface for predicting assay activity from a user-entered SMILES string.

5 Dataset

The Tox21 dataset contains chemical structures and binary assay labels. A label of 1 denotes assay activity and 0 denotes inactivity. Blank values mean that the compound was not measured for that assay; they must not be treated as inactive. The raw CSV contained 7,831 rows. After SMILES validation, 8 invalid structures were removed, leaving 7,823 unique valid molecules.

The project predicts the following assays: NR-AR, NR-AR-LBD, NR-AhR, NR-Aromatase, NR-ER, NR-ER-LBD, NR-PPAR-gamma, SR-ARE, SR-ATAD5, SR-HSE, SR-MMP and SR-p53. These correspond to nuclear receptor (NR) and stress response (SR) pathway assays.

The data are imbalanced. For example, only 2.9% of observed NR-PPAR-gamma labels are active, while SR-ARE has 16.2% active labels. Accuracy alone would be misleading in this setting, which is why ROC-AUC and PR-AUC are reported.

6 Methodology

6.1 Overall Workflow

Figure 1 shows how the project works from a raw molecular structure to a Streamlit prediction.

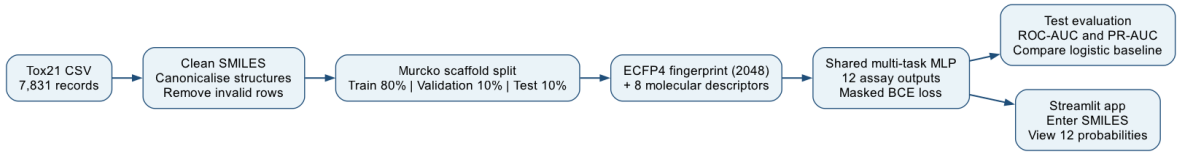


Figure 1: End-to-end workflow of the Tox21 prediction project.

6.2 Data Cleaning

Each input SMILES is checked using RDKit. Valid molecules are converted to canonical SMILES, so equivalent representations are handled consistently. The data-cleaning script also checks duplicate structures. When duplicate structures have conflicting labels for an assay, that assay label is masked rather than forcing an incorrect label.

6.3 Train, Validation and Test Split

The main experiment uses a Bemis–Murcko scaffold split. Molecules having the same core scaffold are kept in only one of the train, validation, or test subsets. The split sizes are 6,258 training compounds (80%), 782 validation compounds (10%), and 783 test compounds (10%). This is stricter than a random split because it evaluates how well the model generalises to new chemical cores.

6.4 Molecular Features

Each molecule is converted into two types of features:

- **ECFP4 fingerprint:** a 2,048-bit circular fingerprint that captures local chemical environments.
- **Simple descriptors:** molecular weight, logP, topological polar surface area, hydrogen-bond donors, hydrogen-bond acceptors, rotatable bonds, ring count, and heavy-atom count.

The descriptor values are standardised using only the training data. The fingerprint and scaled descriptors are joined and passed to the model.

6.5 Multi-Task Neural Network

The proposed model is a shared multi-layer perceptron (MLP). It has an input layer, two hidden layers (256 and 128 units), ReLU activations, dropout, and a final layer with 12 outputs. Each output gives the probability that a molecule is active in one assay.

The loss function is masked binary cross-entropy. If a label is missing, its mask is zero and it contributes no loss. To reduce the effect of class imbalance, each assay has its own positive-class weight. The model was trained with AdamW using a batch size of 128 and learning rate of 0.001. Early stopping selected the best validation mean ROC-AUC; training stopped after 13 epochs.

6.6 Baseline Model

For a fair comparison, 12 separate logistic-regression classifiers were trained. Each baseline model sees the same molecular features, but it learns one assay independently. It uses balanced class weights. This baseline represents a simpler traditional machine learning method without multi-task information sharing.

6.7 Evaluation Metrics

ROC-AUC measures how well active compounds are ranked above inactive compounds over all decision thresholds. PR-AUC is especially informative for rare active compounds because it focuses on precision and recall. Both metrics are calculated on the test set only after model selection using validation data.

7 Implementation

The project is written in Python. RDKit is used for chemical processing, scikit-learn for the baseline model and scaling, PyTorch for the neural network, and Streamlit for the interface. The implementation has a deliberately simple structure:

- `scripts/prepare_data.py`: clean data and create splits.
- `scripts/train.py`: train and evaluate both models.
- `app.py`: Streamlit application for one molecule at a time.
- `src/`: small reusable functions for chemistry, splitting, evaluation, and model definition.

8 Results and Discussion

8.1 Overall Comparison

Table 1 compares both approaches on the held-out scaffold test set.

Table 1: Mean test performance across the 12 assays.

Model	Mean ROC-AUC	Mean PR-AUC
Single-task logistic regression	0.716	0.307
Proposed multi-task MLP	**0.741**	**0.342**

The multi-task MLP improves mean ROC-AUC by 0.025 and mean PR-AUC by 0.035 compared with the baseline. This suggests that shared learning across assays was useful overall. It does not mean that the neural model is better for every assay.

8.2 Per-Assay Performance

Table 2: Test ROC-AUC for each Tox21 assay. Higher is better.

Assay	Logistic regression	Multi-task MLP
NR-AR	0.725	0.717
NR-AR-LBD	0.869	0.826
NR-AhR	0.787	0.830
NR-Aromatase	0.715	0.679
NR-ER	0.611	0.652
NR-ER-LBD	0.715	0.736
NR-PPAR-gamma	0.680	0.769
SR-ARE	0.672	0.686
SR-ATAD5	0.687	0.712
SR-HSE	0.667	0.753
SR-MMP	0.774	0.807
SR-p53	0.693	0.728

The multi-task model performs better on 9 of 12 assays by ROC-AUC. The largest improvement is for NR-PPAR-gamma (0.769 versus 0.680) and SR-HSE (0.753 versus 0.667). The logistic baseline performs better for NR-AR, NR-AR-LBD and NR-Aromatase. This is an important observation: multi-task learning can produce negative transfer for some tasks, so the result should be reported honestly rather than claiming universal improvement.

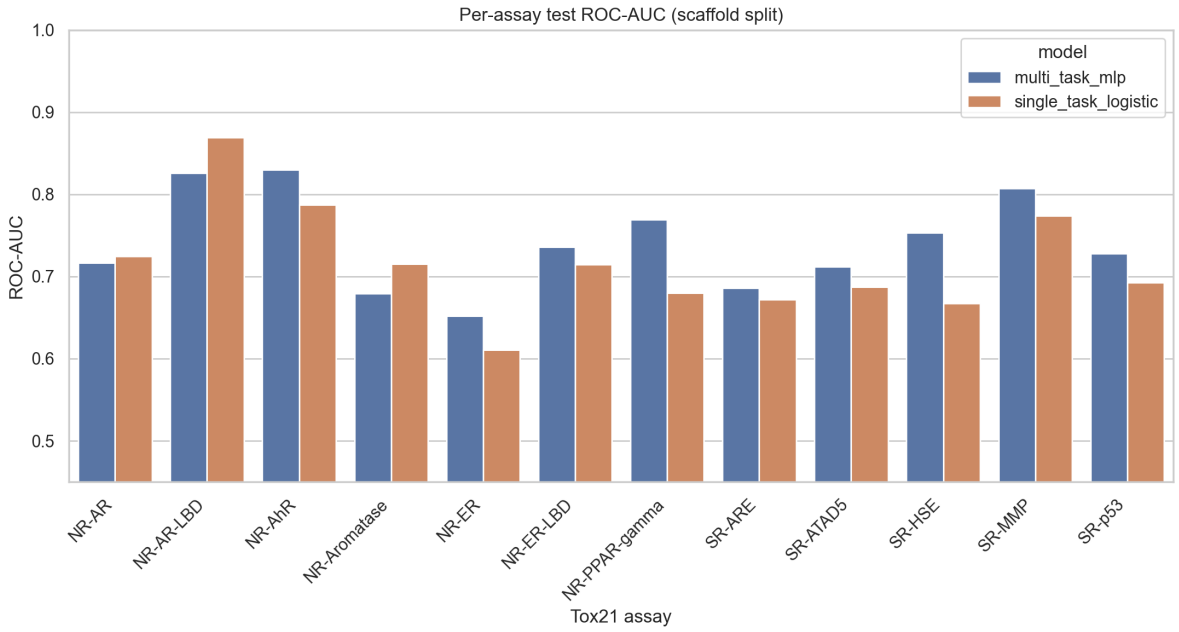


Figure 2: Per-assay scaffold-test ROC-AUC. The chart is generated directly from the saved test metrics.

8.3 Training Behaviour

Figure 3 shows the training loss and validation mean ROC-AUC. The model checkpoint was selected using validation performance, not the test set. This prevents selecting a model by repeatedly checking the final test results.

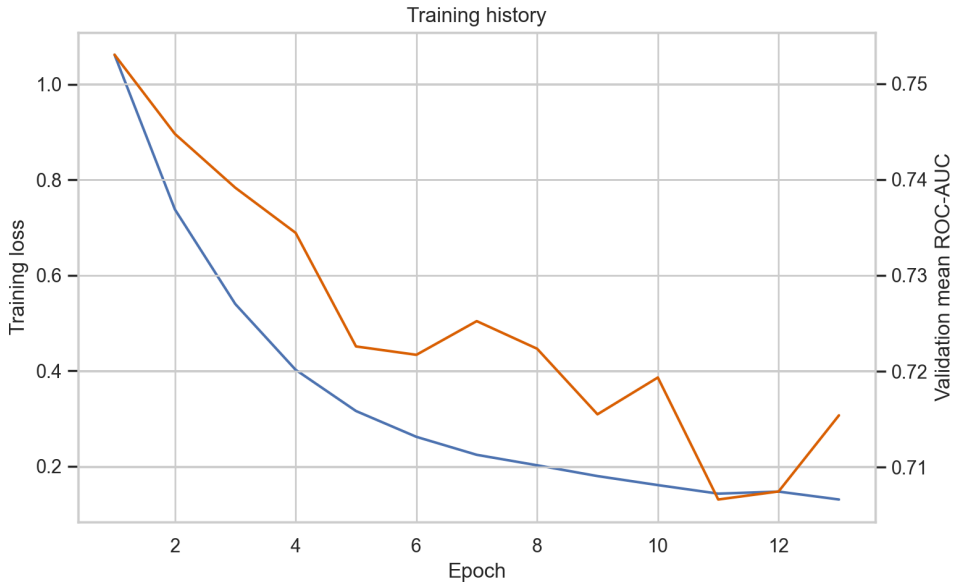


Figure 3: Training history of the multi-task MLP.

9 How the Streamlit Application Works

The Streamlit app accepts one SMILES string. It first validates the structure using RDKit. For a valid molecule, it calculates the same fingerprint and descriptors used in training, applies the saved descriptor scaler, and loads the saved scaffold-trained model. The app then displays:

1. a 2D molecular drawing;
2. predicted activity probabilities for all 12 assays;
3. a bar chart of probabilities;
4. descriptor values; and
5. similarity to the nearest training compound.

The nearest-training similarity is calculated using Tanimoto similarity between ECFP4 fingerprints. Similarity of at least 0.60 is shown as familiar chemical space, 0.35–0.59 as moderate similarity, and below 0.35 as an extrapolation warning. This is a practical warning only; it is not a calibrated uncertainty estimate.

10 Comparison with Traditional or Existing Approaches

The project should not be interpreted as replacing laboratory experiments or as being better than every existing toxicity model. Its measured comparison is limited to the single-task logistic-regression baseline trained in this project on the same scaffold split.

Compared with the traditional baseline, the proposed model has the following advantages:

- **One shared model for 12 assays:** common molecular patterns may be reused between related tasks.
- **Better mean performance in this experiment:** mean ROC-AUC rises from 0.716 to 0.741, and mean PR-AUC rises from 0.307 to 0.342.
- **Correct handling of missing labels:** blank assay results are masked instead of incorrectly converted to inactive labels.
- **Practical accessibility:** the Streamlit application lets a user screen a new SMILES string without retraining the model.
- **More honest evaluation:** scaffold splitting reduces the chance that highly similar compounds are present in both training and test data.

However, the baseline is simpler, faster to train, and better for three assays in this experiment. Therefore, model choice should depend on the intended assay and future validation results.

11 Limitations and Future Work

This work has several limitations:

- Tox21 assay activity is not equivalent to human toxicity, clinical safety, or regulatory risk.
- Labels are sparse, imbalanced, and may include experimental noise.
- The model uses only 2D structure information; dose, metabolism, exposure, and 3D protein interactions are not included.
- Similarity is a simple applicability-domain warning and not a confidence interval.
- Results come from one dataset and one train/test split; external validation is needed.

Future improvements could include graph neural networks, uncertainty calibration, hyperparameter tuning, external validation datasets, and combining toxicity predictions with ADMET and target-activity models.

12 Conclusion

This project demonstrates a complete AI-assisted early toxicity-screening workflow. It converts a molecular SMILES string into numerical chemical features, predicts 12 Tox21 assay activities with a shared neural network, and presents the results in Streamlit. On the scaffold test set, the multi-task model outperformed the simpler single-task logistic baseline in mean ROC-AUC and PR-AUC, while not winning every individual assay. The project is useful as a reproducible educational screening tool and illustrates both the potential and limitations of multi-task learning in drug discovery.

References

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[2] Wu, Z. et al. (2018). MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*, 9, 513–530. https://doi.org/10.1039/C7SC02664A.

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